

Final Project: Incident Response and Forensics

Estimated time: **45** minutes

Welcome to your final project where you will apply your knowledge of incident responses and forensics to a real-world scenario. This is a two-part final project.

In **Part 1**, you'll review a case study and complete related tasks where you identify tasks that are part of the NIST incident response framework for this organization.

In **Part 2**, you will create an incident response plan that outlines the steps your organization would follow for any digital forensics investigation.

Next review the **Before you begin** instructions and follow the instructions to download the files that contains the questions you'll answer as part of your final project. You will also save your answers to these questions in these documents. Later, you'll copy and paste your answers into text files for evaluation and grading.

Before you begin

Before you begin your project tasks, you need following applications available on your computer:

- A text editor application such as:
 - Microsoft Word
 - A PDF viewer
 - Google Docs
 - Notepad or other text editor

To complete your work, you will open the following text-based solution document on your computer and save your answers in that document.

[Final Project Solution document](#)

When you submit your final project answers you will copy and paste your answers from the word document into the provided text upload fields.

Download your documents

This course provides the following text based tasks for Part 1 and Part 2 final project tasks.

1. Right-click to download and save the following word document to an accesible location. You'll enter your answers to the final project questions within this text based document. You can open this document using a Microsoft word, Google Docs, or other compatible word processing program.
2. Next, review the steps needed to save and submit your text-based final project answers.

Saving and submitting your text-based final project answers

Follow the instructions within each file to complete your final project assignments.

1. Locate your downloaded solution file.
2. Follow the instructions within the file to complete your final project assignments.
3. Save your file to an accessible location.
4. Follow the instructions in the course to submit your final project answers.

Let's get started with your final project.

Part 1: Create the incident response plan details using the NIST incident response framework.

Review the following case study. Afterward, open the template, and answer the questions based on the information supplied in the case study. Remember to save your answers in the document provided. You'll copy and paste your answers into text fields located within the online final project assignment.

Case study: Incident response at SecureSync Corporation

SecureSync Corporation is a mid-sized technology company that develops cloud-based solutions for businesses. With a team of over 500 employees, this organization prides itself on innovation and reliability. The company is headquartered in San Francisco with additional offices in Austin and Boston. The company's IT infrastructure supports a client base of approximately 200,000 users. Although the organization feels confident in its incident response capabilities, the company does not have a formalized incident response plan.

Technical details

The company's IT infrastructure follows a hybrid model that combines on-premises servers with cloud-based services, specifically using Azure and Amazon Web Services (AWS) for scalability and redundancy. These services host the company's cloud-based solutions, which are segmented into virtual private networks (VPNs) for enhanced security. The organization's network architecture supports a multilayered security approach, incorporating firewalls, intrusion detection systems (IDS), and data encryption protocols.

Current security tools and protocols

SecureSync Corporation uses the following security tools and protocols:

- **Firewalls:** SecureSync Corp. uses hardware and software firewalls to control incoming and outgoing network traffic based on predetermined security rules. These firewalls are critical in defending against unauthorized access.
- **Intrusion detection and prevention systems (IDPS):** The company uses real-time IDPS to monitor network traffic for suspicious activities and potential threats. These systems alert the IT team when anomalies are detected.

- **Endpoint security:** All employee devices are equipped with endpoint protection software, such as antivirus, threat detection, and response capabilities, to safeguard against malware infection and data breaches.
- **Encryption and data security:** Company data at rest and in transit is encrypted using advanced encryption standards (AES).
- **Access controls:** Role-based access control (RBAC) ensures employees can only access the resources necessary for their roles, minimizing the risk of internal threats.
- **Logging and monitoring:** Comprehensive logging configurations capture detailed records across all systems and applications. These logs play a crucial role in incident detection and retrospective analysis.

Although SecureSync Corp. understands the importance of incident response, the company does not have a formal incident response plan documented. The Chief Information Officer (CIO) recognizes the need to document the preparedness plan, which will now include:

- Updated staff training programs focused on recognizing phishing attempts,
- Prompt reporting of suspicious activity
- Regular security drills.

The organization has established key personnel roles, which include an incident response team leader, a communication liaison, and technical specialists.

The organization also invested in advanced detection software systems to enhance their defenses against cyberthreats.

Incident response in action

On a busy Monday morning, the IT security team at SecureSync Corp. detected unusual network activity. The IT security team identified multiple unauthorized access attempts on the company's email server, which stores sensitive client data and internal communications. This attempt raised concerns about a potential data breach. Initial scans revealed malware deployed to siphon sensitive data to an unknown offsite server.

During the incident investigation, the following information was discovered:

- The IT security team received alerts about anomalous behavior indicative of a cyberattack using the company's Incident Detection System (IDS).
- Log files revealed remote logins occurring at odd hours and from unusual locations.
- Automated scripts captured packet details, suggesting potential data exfiltration.

The company activated the incident response team upon confirmation of the breach and began a detailed investigation. Analysts discovered that the attack vector was a sophisticated spear-phishing email that compromised user email accounts and gave attackers backdoor access. The incident response team constructed a timeline to trace the incident's development and identify affected systems.

The team's rapid response initially focused on short-term containment to limit further damage. The team sealed off critical access points by deactivating compromised accounts and isolating affected network segments. The company performed an extensive threat hunt to identify the malware. Threat intelligence research helped the team understand the malware's typical behavior. The team used these insights to devise an eradication plan that included removing malicious code from all infected systems. After confirming malware eradication, the team shifted their efforts to recovery. The team restored systems from secure backups and conducted system-wide checks to ensure no residual damage or vulnerabilities remained undetected. The team then established continuous monitoring to detect any attempts at further breaches or compromises.

With the incident under control, SecureSync Corp. initiated a comprehensive post-mortem analysis to evaluate its incident response strategy. The analysis focused on response times, communication efficiency, security protocol effectiveness, and areas needing improvement. Team debriefings highlighted necessary enhancements in the incident response plan, particularly regarding threat intelligence sharing and expedited cross-departmental communication. New preventive measures were discussed and documented, and actions were taken to patch identified vulnerabilities.

SecureSync Corp. also planned educational workshops to reinforce cybersecurity practices among employees to mitigate potential future threats. The management also recognized the importance of public transparency and issued a client-facing report assuring clients of improved security measures, reinforcing trust, and retaining brand credibility.

Next, complete your **Part 1** final project assignment tasks.

Complete the Final Project Part 1 Assignment Tasks

After you open the **Part 1 solution document** where you can save your answers complete the following tasks. The following tasks are numbered to match how you will supply the answers for **Submission and Evaluation**

Prompt 1

List the roles and responsibilities of four key team members in the incident response team.

Prompt 2

List the four methods, or tasks, you can use to monitor this company's internal systems for unusual activity.

Prompt 3

Using what you know about the NIST framework and this company, list four details to include when documenting detected incidents.

Prompt 4

List at least two containment strategies and explain how these strategies will help the company with containment.

Use the following format:

Containment strategy name: A sentence that explains how the strategy will help the company with containment.

Prompt 5

List the four-steps the company would use to conduct post-incident reviews based on the NIST framework.

List the answers using the following format:

Step 1: List the first step here
Step 2: List the second step here
Step 3 : List the third step here
Step 4 : List the fourth step here

Prompt 6

Using the NIST framework, write a checklist of three tasks an organization can use to structure an approach for updating the response plan based on findings.

Prompt 7

Based on the case study and the NIST framework, list four sources of digital evidence necessary for incident investigation.

Write your answers in a list using the following format:

Source 1: List the first source of digital evidence.
Source 2: List the second source of digital evidence.
Source 3: List the third source of digital evidence.
Source 2: List the fourth source of digital evidence.

Prompt 8

List the three steps required to assess the collected digital evidence and verify its integrity.

Use the following format:

Step 1: List the step information here.
Step 2: List the step information here.
Step 3: List the step information here.

2. Validate that you completed all of your answers for prompts 1 - 8.
3. If you did not already renamed your text document to use it later, rename the Part 1 template document as ***Solution document_YourInitials***, and save the document in an accessible location for later use.
4. Begin your work for Part 2.

Part 2: Forensic investigations: Creating an Incident Response Plan

In part 2 of this final project, you will document steps and reasoning for an incident response plan that outlines the steps an organization would follow for any digital forensics investigation.

Your company will use this document as its standardized process and checklist to ensure consistency, thoroughness, and integrity in future investigations. You will divide the plan into four phases, each with specific objectives.

Prompt 9

List the three types of digital evidence that the organization should review as part of a forensic investigation to determine the breach's origin and method. Then explain the purpose for that digital evidence.

Provide your answer using the following format:

Evidence Type 1: Write a sentence that explains the purpose for this digital evidence.
Evidence Type 2: Write a sentence that explains the purpose for this digital evidence.
Evidence Type 3: Write a sentence that explains the purpose for this digital evidence.

Prompt 10

Using what you know about digital forensics, list four key components to include in structured reports following each incident and describe each component.

Provide your answer using the following format:

First key component: Describe this component in one or two sentences.
Second key component: Describe this component in one or two sentences.
Third key component: Describe this component in one or two sentences.
Fourth key component: Describe this component in one or two sentences.

Congratulations! You've now completed all final project assignments!

2. Validate that you completed all sections.
3. Save your answers into your ***Solution document_YourInitials***, and save the document in an accessible location for later final project submission.

Finalize your assignment

Before you submit your assignment, complete these steps:

1. Verify that you provided answers to all tasks.
2. Remember to rename the solution document as described (Project solution document_YourInitials)
3. Remove the instruction sections or examples provided in the Project solution document. Your final deliverables should only contain your work. However, keep the answers that were provided for you.

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